

1 Fundamental Concepts

1.1 General useful relations

Waves & Photons λ [m] = $\frac{c}{\nu} = \frac{2\pi c}{\omega} = \frac{2\pi}{k}$ ν [Hz] = $\frac{c}{\lambda}$
 ω [rad s⁻¹] = $2\pi\nu = ck$ k [m⁻¹] = $\frac{2\pi}{\lambda} = \frac{\omega}{c}$ E [J] = $h\nu = \hbar\omega$
 $h\nu = \frac{hc}{\lambda}$ p [kg m s⁻¹] = $\frac{h}{\lambda} = \hbar k$ h [J s], $\hbar$ [J s] = $\frac{h}{2\pi}$

Solid Angle $\Omega = \frac{A}{R^2}$ $\Omega_{\text{sphere}} = 4\pi$ sr

Angle Conversion 1° = 1 deg = 1 · $\frac{180}{\pi}$ rad
1 deg = 1 · $(\frac{180}{\pi})^2$ sr | 1' = $\frac{180 \cdot 60}{\pi}$ rad | 1'' = $\frac{180 \cdot 3600}{\pi}$ rad

1.2 Astronomic Orientation

Parallax: As the Earth or **Galactic coordinate systems** the Sun, the apparent **tem:** Origin at the Sun; position of a star at distance d ; for small angles $\theta \simeq \frac{1 \text{ AU}}{d}$.
Galactic latitude $-90^\circ \leq b \leq 90^\circ$ (Galactic poles at $b = \pm 90^\circ$) and Galactic longitude $0^\circ \leq l \leq 360^\circ$. Galactic center $l = b = 0^\circ$

1.3 Luminosity, Magnitudes & Flux

Luminosity (intrinsic): L [erg s⁻¹], (total/bolometric) **Flux** (observed at distance d): F [erg s⁻¹ cm⁻²] (1 erg = 10⁻⁷ J).
 $F = \frac{L}{4\pi d^2}$.

Magnitudes: Apparent magnitude m measures the observed brightness (flux). Absolute magnitude M is the magnitude the object would have at $d = 10$ pc (intrinsic luminosity measure).
For two sources with fluxes F_1, F_2 and apparent magnitudes m_1, m_2 :

$$m_1 - m_2 = \frac{5}{2} \log_{10} \left(\frac{F_2}{F_1} \right)$$

Distance modulus: $m - M = 5 \log_{10} \left(\frac{d}{10 \text{ pc}} \right) + \alpha_\nu$

Color index: difference between magnitudes in two passbands, e.g. $B - V$; larger for redder sources and smaller for bluer sources. Distance and brightness differences cannot be inferred by difference in colors alone.
Bolometric magnitude: flux integrated over all λ .

1.4 Gravitation

Classical gravity (Newton): Gravity is described as an action at a distance, $F_G = G \frac{Mm}{r^2}$.

Grav. binding energy (extended sphere):
 $U = - \int_0^R \frac{GM(r)}{r} dm$, $dm = 4\pi r^2 \rho(r) dr$, $M(r) = \int_0^r 4\pi r'^2 \rho(r') dr'$ ($\rho = \text{const}$) $\Rightarrow U = - \frac{3GM^2}{5R}$

General Relativity (Einstein): Gravity is not a force but a manifestation of spacetime curvature caused by energy density. The kinetic and gravitational potential energies of a particle of mass m are

$$K = \frac{1}{2}mv^2, \quad U = - \frac{GMm}{r},$$

with total energy

$$E = K + U = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

A particle is gravitationally bound if $E < 0$.

Heuristic estimate (apply to a photon, $v \simeq c$):
 $E < 0 \Rightarrow r < \frac{2GM}{v^2} \xrightarrow{v=c} r < \frac{2GM}{c^2}$.

This defines the **Schwarzschild radius** (a characteristic relativistic length scale),

$$r_s := \frac{2GM}{c^2}$$

Defining the compactness parameter

$$f = \frac{r_s}{r} = \frac{2GM}{rc^2}$$

General Relativity becomes important if $f \gtrsim 1$, i.e. if $r \lesssim r_s$.
For an object of constant density ρ and radius r , the mass is $M = \frac{4}{3}\pi r^3 \rho$, which yields $f = \frac{8\pi G \rho r^2}{3c^2}$.

1.5 Sources of Astrophysical Information

Astrophysical observations rely on several sources of information: electromagnetic radiation, neutrinos (very weak interaction with matter via the weak force), gravitational waves, cosmic rays (high-energy particles e.g. electrons, protons, nuclei).

Earth's atmosphere is transparent mainly to *visible* and *radio wavelengths*. Observations at other wavelengths (UV, IR, X-ray, Gamma) require space-based telescopes.

1.6 Visible Astronomy: Telescopes

There are two main types of telescopes: **refractive optics** (glass lenses; mainly historical) and **reflecting optics** (mirrors; standard today). Reflectors are free from chromatic effects, and mirrors are easier to manufacture and support mechanically.

Telescope diameter D : diameter of the primary optical element (main light-collecting component). It sets

$A \propto D^2$ (light-gathering power), $\theta \propto D^{-1}$ (angular resolution).

Angular resolution: minimum angular separation θ that can be resolved by a diffraction-limited circular aperture:

$$\theta_{\text{diff}} \simeq 1.22 \frac{\lambda}{D} \quad [\text{rad}]$$

(Rayleigh criterion for a single-aperture diffraction pattern). Atmospheric seeing limits the angular resolution to $\sim 0.5'' \approx 2.4 \times 10^{-6}$ rad.

2 Interaction of Radiation with Matter

Two levels: **macroscopic** (radiative transfer) and **microscopic** (emission/absorption).

2.1 Radiative Transfer: General (Radiation Field)

Specific intensity $I_\nu(\vec{r}, t, \hat{n})$:
 $J_{\hat{n}}$: energy arriving from direction $\hat{n}$; $m_\perp^2$: area perpendicular to ray; s : time; sr : solid angle; Hz: per frequency.

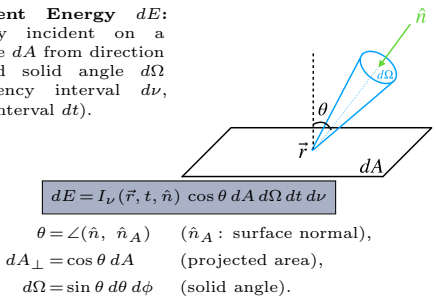
$$\left[\frac{J_{\hat{n}}}{m_\perp^2 \text{ s sr Hz}} = \frac{\text{W}}{\text{m}^2 \text{ sr Hz}} \right]$$

Solid angle:

$$\int_{4\pi} d\Omega = \int_0^{2\pi} \int_0^\pi \sin \theta \, d\theta \, d\phi = 4\pi.$$

4π corresponds to integration over all directions (full sphere); a half sphere ($\theta \in [0, \pi/2]$) gives 2π .

Incident Energy dE :
Energy incident on a surface dA from direction $\hat{n}$ and solid angle $d\Omega$ (frequency interval $d\nu$, time interval dt).



Specific Flux F_ν : Net energy through a (unit) area per unit time and per unit frequency, integrated over all directions. If you integrate specific flux over all frequencies, you get the **total (bolometric) flux** $F = \int F_\nu \, d\nu$, with $[W/m^2]$.

$$F_\nu = \int_{\text{All dir.}} \frac{dE}{dA \, dt \, d\nu} = \int_{4\pi} I_\nu \cos \theta \, d\Omega \quad \left[\frac{\text{W}}{\text{m}^2 \text{ Hz}} \right]$$

Energy density u_ν : energy of radiation (with frequency ν per volume): Cylinder with length ds and cross-sectional area $dA \perp \hat{n}$.

$$du_\nu \equiv \tilde{u}_\nu(\hat{n}) \equiv \frac{dE}{dV \, d\Omega \, d\nu}$$
$$(dE = I_\nu \, dA \, d\Omega \, dt \, d\nu), \quad (dV = dA \, ds)$$
$$\Rightarrow du_\nu = \frac{I_\nu \cancel{dA} \, d\Omega \, dt \, d\nu}{\cancel{dA} \, ds \, d\nu}, \quad (dt = \frac{ds}{c})$$
$$\Rightarrow du_\nu = \tilde{u}_\nu(\hat{n}) = \frac{I_\nu}{c} \, d\Omega$$

$$u_\nu = \int_{4\pi} \tilde{u}_\nu(\hat{n}) \, d\Omega = \frac{1}{c} \int_{4\pi} I_\nu \, d\Omega \quad \left[\frac{\text{J}}{\text{m}^3 \text{ Hz}} \right]$$

Radiation pressure P_ν : (momentum flux per area per frequency; momentum component normal to surface)

$$p_\gamma = \frac{E}{c}, \quad E = h\nu, \quad p_\perp = p_\gamma \cos \theta$$
$$dp_\perp = \frac{dE}{c} \cos \theta = \frac{I_\nu}{c} \cos^2 \theta \, dA \, d\Omega \, dt \, d\nu$$
$$P_\nu = \frac{1}{c} \frac{dE}{dA \, dt \, d\nu} \int_{4\pi} dp_\perp = \frac{1}{c} \int_{4\pi} I_\nu \cos^2 \theta \, d\Omega$$

$$P_\nu = \frac{1}{c} \int I_\nu \cos^2 \theta \, d\Omega \quad \left[\frac{\text{N}}{\text{m}^2 \text{ Hz}} = \frac{\text{Pa}}{\text{Hz}} \right]$$

2.2 Radiative Transfer: Special Cases

Special case: Isotropic radiation field For an isotropic radiation field,

$$I_\nu(\vec{r}, t, \hat{n}) = I_\nu(\vec{r}, t),$$

i.e. the intensity is independent of direction $\hat{n}$. Hence I_ν can be taken outside the angular integrals (because it is now a constant factor). Using $\int_{4\pi} d\Omega = \int_0^{2\pi} \int_0^\pi \sin \theta \, d\theta \, d\phi$, the moments of the intensity simplify to:

$$F_\nu = \int I_\nu \cos \theta \, d\Omega = 0 \quad U_\nu = \frac{1}{c} \int I_\nu \, d\Omega = \frac{4\pi}{c} I_\nu$$
$$P_\nu = \frac{1}{c} \int I_\nu \cos^2 \theta \, d\Omega = \frac{1}{3} \frac{4\pi}{c} I_\nu = \frac{1}{3} U_\nu$$

2.3 Blackbody Radiation

Blackbody radiation: Radiation in a cavity in thermal equilibrium at temperature T . The radiation field is isotropic, hence

$$U_\nu = \frac{4\pi}{c} I_\nu, \quad I_\nu \equiv B_\nu(T).$$

Planck's law: Spectral radiance of a blackbody:

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \quad \text{or}$$
$$B_\lambda(T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1}$$
$$[B_\nu] = \frac{\text{W}}{\text{m}^2 \text{ sr Hz}} \quad [B_\lambda] = \frac{\text{W}}{\text{m}^3 \text{ sr}}$$

Approximations to Planck's law (limits):

$h\nu \ll k_B T$: Rayleigh-Jeans (long wavelengths, low frequencies)
 $h\nu \gg k_B T$: Wien (short wavelengths, high frequencies)

$$B_\nu(T) \simeq \begin{cases} \frac{2\nu^2 k_B T}{c^2}, & h\nu \ll k_B T \quad (\text{R.J}) \\ \frac{2h\nu^3}{c^2} e^{-h\nu/k_B T}, & h\nu \gg k_B T \quad (\text{Wien}) \end{cases}$$

Wien's displacement law: Wavelength of maximum emission: $\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m K}$.

Stefan-Boltzmann law: Total power radiated per unit area ($F_\nu = c U_\nu / 4 = \pi B_\nu$) (Hemisphere), $a_B = 4\sigma/c$:

$$F = \int_0^\infty F_\nu \, d\nu = \sigma T^4, \quad \sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}.$$

Total luminosity of a spherical source: $L = 4\pi R^2 F$. $\left[\frac{\text{W}}{\text{m}^2} \right]$

2.4 Radiative Transfer Equation & Solutions

Radiative transfer equation (RTE):

$$\frac{dI_\nu}{ds} = j_\nu - \alpha_\nu I_\nu$$
$$[\alpha_\nu] = \text{m}^{-1}, \quad [j_\nu] = \frac{\text{W}}{\text{m}^3 \text{ sr Hz}}.$$

α_ν is the absorption coefficient, j_ν is the emissivity. There are three cases for the radiative transfer equation:

Empty space: $\alpha_\nu = 0$, $j_\nu = 0$ Empty space neither emits nor absorbs radiation:

$$\frac{dI_\nu}{ds} = 0 \Rightarrow I_\nu = \text{const.}$$

Thus, I_ν is conserved along a ray. As the distance R increases, both the flux F_ν and the solid angle Ω decrease, compensating each other:

$$I_\nu = \frac{F_\nu}{\Omega} = \frac{L_\nu}{4\pi R^2} \frac{R^2}{A} = \text{const.}$$

Absorption only: $\alpha_\nu \neq 0$, $j_\nu = 0$ Radiation is attenuated along the path:

$$\frac{dI_\nu}{ds} = -\alpha_\nu(s) I_\nu \Rightarrow \int_{I_\nu(s_0)}^{I_\nu(s)} \frac{dI_\nu}{I_\nu} = - \int_{s_0}^s \alpha_\nu(s') \, ds'$$
$$\ln \left(\frac{I_\nu(s)}{I_\nu(s_0)} \right) = - \int_{s_0}^s \alpha_\nu(s') \, ds'$$

$$I_\nu(s) = I_\nu(s_0) \exp \left[- \int_{s_0}^s \alpha_\nu(s') \, ds' \right]$$

Optical depth τ_ν : Dimensionless measure of the opacity of matter. — ($\alpha_\nu = \rho \chi$), χ can depend on ν .

$\tau_\nu \gg 1$: optically thick $\rightarrow$ opaque (lot's of absorption)

$\tau_\nu \ll 1$: optically thin $\rightarrow$ transparent (not much absorption)

$$d\tau_\nu = \alpha_\nu \, ds, \Rightarrow \tau_\nu = \int \alpha_\nu(s) \, ds$$

$$I_\nu(s) = I_\nu(s_0) e^{-\tau_\nu}$$

General Case: $\alpha_\nu \neq 0$, $j_\nu \neq 0$

$$\text{Source function: } S_\nu \equiv \frac{j_\nu}{\alpha_\nu}$$

Divide radiative transfer equation by α_ν :

$$\frac{dI_\nu}{ds} = S_\nu - I_\nu \frac{\alpha_\nu ds = d\tau_\nu}{d\tau_\nu} \Rightarrow \frac{dI_\nu}{d\tau_\nu} = S_\nu - I_\nu,$$

$$\frac{d}{d\tau_\nu} (I_\nu e^{\tau_\nu}) = \frac{dI_\nu}{d\tau_\nu} e^{\tau_\nu} + I_\nu e^{\tau_\nu} = (S_\nu - I_\nu) e^{\tau_\nu} + I_\nu e^{\tau_\nu}$$

$$= S_\nu e^{\tau_\nu},$$

$$\int_0^{\tau_\nu} \frac{d}{d\tau_\nu'} \left(I_\nu e^{\tau_\nu'} \right) d\tau_\nu' = \left[I_\nu(\tau_\nu') e^{\tau_\nu'} \right]_0^{\tau_\nu} = \int_0^{\tau_\nu} S_\nu(\tau_\nu') e^{\tau_\nu'} d\tau_\nu'$$

$$\Rightarrow I_\nu(\tau_\nu) e^{\tau_\nu} - I_\nu(0) = \int_0^{\tau_\nu} S_\nu(\tau_\nu') e^{\tau_\nu'} d\tau_\nu'$$

$$I_\nu(\tau_\nu) = I_\nu(0) e^{-\tau_\nu} + \int_0^{\tau_\nu} e^{-(\tau_\nu - \tau_\nu')} S_\nu(\tau_\nu') d\tau_\nu'$$

Matter with constant properties: $S_\nu(\tau_\nu) = S_\nu$, $\alpha_\nu(\tau_\nu) = \alpha_\nu$, both S_ν and α_ν are constant. No background source $I_\nu(0) = 0$. Factorize out the constants, general solution of RTE reduces to:

$$I_\nu(\tau_\nu) = S_\nu (1 - e^{-\tau_\nu})$$

$\tau_\nu \gg 1$: optically thick $\rightarrow I_\nu \simeq S_\nu$

$\tau_\nu \ll 1$: optically thin $\rightarrow I_\nu \simeq S_\nu \tau_\nu = j_\nu L$

Kirchhoff's Law: optically thick objects in LTE (local thermal equilibrium), $B_\nu = \text{Blackbody Radiation}$:

$$I_\nu = B_\nu(T), \quad j_\nu = \alpha_\nu B_\nu(T)$$

2.5 Radiative Transfer for Stars

Assumptions: steady state, locally curvature negligible, LTE, optically thick.

$$dz = \cos \theta \, ds, \quad I_\nu(\vec{r}, \hat{n}) = I_\nu(z, \theta), \quad T = T(z)$$

Using $S_\nu = j_\nu / \alpha_\nu$ and $ds = dz / \cos \theta$ in the RTE:

$$\frac{\cos \theta}{\alpha_\nu} \frac{dI_\nu}{dz} = S_\nu - I_\nu$$

LTE & optically thick $\Rightarrow S_\nu = B_\nu(T)$:

$$I_\nu = B_\nu - \frac{\cos \theta}{\alpha_\nu} \frac{dI_\nu}{dz}$$

Using I_ν you can calculate **Eddington approximation** (optically thick):

$$F_\nu = - \frac{c}{\alpha_\nu} \frac{dP_\nu}{dz} \quad P_\nu \simeq \frac{4\pi}{3c} B_\nu$$

Rosslands mean absorption coefficient α_R :

$$\frac{1}{\alpha_R} = \frac{\int \frac{1}{\alpha_\nu} \frac{dB_\nu}{dT} d\nu}{\int \frac{dB_\nu}{dT} d\nu}$$

Using the Stefan-Boltzmann law and the mass opacity $\chi = \alpha_R / \rho$, the bolometric flux reads:

$$F = - \frac{4\pi}{3} \frac{dT}{dz} \frac{1}{\alpha_R} \frac{d}{dT} \int B_\nu d\nu = - \frac{c}{\chi \rho} \frac{d}{dz} \left(\frac{g}{3} T^4 \right)$$

$$\alpha_\nu: \text{absorption coeff.} \quad \alpha_\nu = -\frac{1}{I_\nu} \frac{dI_\nu}{ds} \Big|_{\text{abs}}, [\alpha_\nu] = \text{m}^{-1}$$

$$\tau_\nu: \text{optical depth} \quad d\tau_\nu = \alpha_\nu ds, [\tau_\nu] = 1$$

$$l_\nu: \text{mean free path} \quad l_\nu = 1/\alpha_\nu, [l_\nu] = \text{m}$$

$$\alpha_R: \text{Rosseland mean} \quad [\alpha_R] = \text{m}^{-1}$$

$$\chi: \text{mass opacity} \quad \chi = \alpha_R/\rho, [\chi] = \text{m}^2 \text{ kg}^{-1}$$

$$\alpha: (\text{bol.}) \text{ abs. coeff.} \quad \alpha = n\sigma, [\alpha] = \text{m}^{-1}$$

$$l_\nu: \text{mean free path} \quad 1/\alpha_\nu, [l_\nu] = \text{m}$$

$$n: \text{number density} \quad N/V, [l_\nu] = \text{m}^{-3}$$

2.6 Thermodynamic Equilibrium

Maxwellian distribution

For a two-level system of particles in thermal equilibrium with number densities (Besetzungswahrscheinlichkeiten) n_0 (ground state) and n_e (excited state), the Boltzmann distribution reads:

$$\frac{n_e}{n_0} = e^{-\frac{\Delta E}{k_B T}}.$$

Thermal equilibrium (TE) Achieved through collisions when the mean free path is much smaller than the macroscopic scale:

$$l_\nu \ll L_{\text{sys}} \quad (\text{"local"} \text{ TE})$$

Smaller $l_\nu \Rightarrow$ more frequent interactions $\Rightarrow$ faster relaxation to TE.

Example (sunlight in a room): air molecules have very small l_ν and quickly equilibrate among themselves, but photons interact only weakly with the air (very large photon mean free path), so the radiation field is not in TE with the gas (unless the medium is strongly absorbing/ionised).

3 Stars

3.1 Stellar Structure

Star is spherically approximated. $\Leftarrow$ real imperfections: Rotation, Magnetic Field

Hydrostatic Equilibrium: Star conserves mass and its pressure balances gravity.

$$\frac{dM_r}{dr} = 4\pi r^2 \rho \quad \left| \quad \frac{dP}{dr} = -\frac{GM_r \rho}{r^2} \right| \quad dP \, dA = -\frac{GM_r \rho}{r^2} \, dr \, dA$$

$$\begin{array}{l} \textbf{Scales Sun} \mid M_\odot \simeq 1.99 \times 10^{30} \text{ kg} \mid L_\odot \simeq 4 \times 10^{26} \text{ W} \\ \mid R_\odot \simeq 7 \times 10^8 \text{ m} \mid P_\odot \simeq 6 \times 10^{13} \text{ N/m}^2 \mid T_\text{c} \simeq 10^7 \text{ K} \\ \text{Temperature from: } P = nk_B T \text{ with } n = 2\rho/m_H \end{array}$$

3.2 Energy Transport

Radiative Transfer, Convection, Conduction (Conduction only for white dwarfs)

Energy Generation:

$$\text{Luminosity @ } r = \frac{\text{Energy @ } r}{\text{Mass} \times \text{Time}} \implies dL_r = \rho c dV = \rho c 4\pi r^2 dr$$

Stars generate energy in their core. The energy is transported outwards mainly by radiation and convection.

$$\left. \frac{dL_r}{dr} = 4\pi r^2 \epsilon \right| \quad \epsilon = \text{energy generation term}$$

Radiative Transport

L_r can be expressed with the Flux F: $L_r = 4\pi r^2 F$.

$$\left. \frac{dT}{dr} = -\frac{3}{4\sigma c} \frac{\chi \rho}{T^3} \frac{L_r}{4\pi r^2} \right| \quad L_r = -4\pi r^2 \frac{c}{\chi \rho} \frac{d}{dr} \left(\frac{\sigma T^4}{3} \right)$$

Convective Transport

Acoustic waves ($c_s = \sqrt{\gamma P/\rho}$) **equilibrate pressure fast**. Adiabatic displacement of a blob $P\rho^{-\gamma} \propto \text{const}$ with $\gamma = 5/3$. If the **radiation gradient** cools faster than the **adiabatic cooling gradient**, the blob can continue to rise/cool $\Rightarrow$ Current layering/transport is **unstable, convection happens**.

Temperature	Density	Convection?
$\nabla_{\text{rad}} < \nabla_{\text{adi}}$	$\rho^* > \rho_{\text{crit}}$	No/stable
$\nabla_{\text{rad}} > \nabla_{\text{adi}}$	$\rho^* < \rho_{\text{crit}}$	Yes/unstable

When convection can happen

(unstable layering) the **convective transport** is equal to the adiabatic gradient.

$$\left. \frac{dT}{dr} \simeq \left(1 - \frac{1}{\gamma}\right) \frac{T}{P} \frac{dP}{dr} \right|$$

3.3 Stellar Model

A stellar model can happen with 7 coupled equations for 7 variables: $M_r, L_r, P, T, \rho, \epsilon, \chi$. Boundary conditions: $r=0$: $M_r = 0, L_r = 0$ and $r=R$: $\rho = 0, T = 0$.

Mass continuity, Hydrostatic equilibrium, Energy generation, Energy tranport, Opacity χ , Energy generation rate ϵ . Equation of State.

Equation of State $\frac{1}{\rho} = \frac{1}{n\mu m_H}$
Particle density: $n = \left(2X + \frac{3}{4}Y + \frac{1}{2}Z\right) \frac{\rho}{m_H} = \frac{\rho}{\mu m_H}$, where X, Y, Z are the mass fractions of H, He, and metals.

$$P = nk_B T = \frac{\rho k_B T}{\mu m_H}$$

For very compact stars we need degenerate pressure. $\mu_{\text{H I}} = 1$, (neutral), $\mu_{\text{H II}} = 0.5$ (ionized)

3.4 Scaling Relations

From the stellar structure equations, scaling equations can be derived. Approximate derivatives with variables: $\frac{dX}{dr} \sim \frac{X}{R}$. Plug in other scaling relations.

Stellar Structure Scaling Relations

Mass Continuity	$\frac{P}{R} \propto \frac{M}{R^2} \rho \implies P \propto \frac{M^2}{R^4}$
Equation of State	$P \propto \rho T \implies T \propto \frac{M}{R}$
Radiative Transport	$\frac{T}{R} \propto \frac{\kappa \rho}{T^3} \frac{L}{R^2} \implies L \propto M^3 \quad \text{Exp. } M^{3.7}$
Stellar Surface	$L \propto R^2 T_{\text{eff}}^4 \implies L \propto T_{\text{eff}}^6 \quad \text{Exp. } T^{5.6}$
Lifetime	$L \propto \frac{\text{Energy}}{t} \implies \tau \propto M^{-2}$

Other Scaling Relations: $M \propto \text{Energy}, R \propto M^{0.75}$

3.5 Stellar Observation

From measuring **frequency-dependent flux** F_ν , **frequency** ν and **apparent magnitude** m ; the **temperature** (BlackBody), the **chemical composition** (spectral lines), the **magnetic fields** (Zeeman) and **velocities** (Doppler) are inferred.

3.6 Main Sequence

Luminosity vs. Bolometric Magnitude HR-diagram, Main Sequence: phase of core H $\rightarrow$ He fusion where stars spend most of their lifetime. After core H exhaustion stars leave the main sequence, (Lower surface Temp., higher Lumin.). **Bolometric magnitude** varies oppositely with **temperature**.

Lower Limit $M < 0.08 \cdot M_\odot \Leftarrow$ Brown Dwarfs
Too small star masses cannot initiate fusion process.

Upper Limit

Radiation pressure vs gravity for big stars. $M > 100 \cdot M_\odot$

$$\underbrace{\frac{GM}{R^2}}_{\text{gravity}} \rho = \underbrace{\left[\frac{dP}{dr}\right]}_{\text{pressure}} = \frac{\alpha}{c} F = \frac{\chi \rho}{4\pi R^2} L.$$

$$\text{Maximal Luminosity: } L_{\text{Edd}} = \frac{4\pi c G M}{\chi}$$

From scaling relation $L \propto M^3$ and L_{Edd} a maximal stable mass $M_{\text{max}} \simeq 100 M_\odot$ is derived. $M_{\text{max}} = \left(\frac{4\pi c G}{\chi \lambda}\right)^{1/2}$ with λ from $L = \lambda M^3$

Star-Clusters Main Sequence fitting can give information about the distance and age of stars and globular clusters.

- Open cluster: loosely bound, $\leq 10^2$ young stars
- Globular clusters: tightly bound, $\sim 10^5$ stars
- Star within a star cluster were born at about the same time.

3.7 Nuclear Energy Production

$$E_{\text{grav}} \rightarrow E_{\text{Heat}}: \tau_{\text{KH, Sun}} \approx 10^7 \text{ yrs} \Rightarrow \text{too short}$$

Nuclear Binding Energy

$$E_B = [Zm_p + (A - Z)m_n - m_{\text{nuc}}] c^2 \geq 0$$

Binding energy per nucleon: $f = \frac{E_B}{A}$. Max at $^{56}\text{Fe/Ni} \Rightarrow$ Fusion for light nuclei, Fission for heavy nuclei.

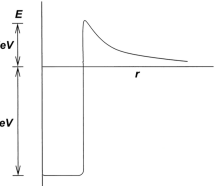
Stellar Lifetime Star converting H $\rightarrow$ He: Fraction of mass converted to energy $\times$ Total mass available = Energy released. (Upper Limit, actual $\tau_{\text{sun}} \approx 10^{10}$ yrs.)

$$\tau_{\text{nuc}} = 0.007 m_H c^2 \frac{M_\odot}{m_H} \frac{1}{L_\odot} = \frac{0.007 M_\odot c^2}{L_\odot} \approx 10^{11} \text{ yrs.}$$

Nuclear Potential

When fusing 2 nuclei the **Coulomb Force** competes with **Strong Nuclear Force**.

Sun: $T_c \approx 1.5 \cdot 10^7 \text{ K} \implies k_B T \approx 1 \text{keV} \ll Z_1 Z_2 \text{MeV} \implies$ tunneling effect needed.



Energy Generation

Energy generation rate: $\rho \epsilon = r \Delta E$ with ΔE energy released per reaction.

Nuclear reaction rate per volume: $r = n_1 n_2 \langle \sigma v \rangle$ with crosssection σ , rel. velocity v , particle densities n_1, n_2 .

PPI-Branch Dominates in Sun $\approx 83.3\%$

$$\begin{array}{l} \text{(I)} \quad {}^1\text{H} + {}^1\text{H} \rightarrow {}^2\text{H} + e^+ + \nu_e \\ \text{(II)} \quad {}^2\text{H} + {}^1\text{H} \rightarrow {}^3\text{He} + \gamma \\ \text{(III)} \quad {}^3\text{He} + {}^3\text{He} \rightarrow {}^4\text{He} + 2 \cdot {}^1\text{H} \\ \text{Overall: } 4 \cdot {}^1\text{H} \rightarrow {}^4\text{He} + 2e^+ + 2\nu_e + \gamma \\ E_{\text{PP1}} = 26.2 \text{ MeV} = 4.20 \times 10^{-12} \text{ J}, \eta = 0.7\% \end{array}$$

PPII-Branch 16.68 % in Sun.

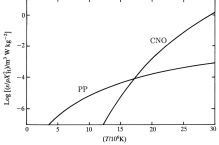
$$\begin{array}{l} \text{(IV)} \quad {}^7\text{Be} + e^- \rightarrow {}^7\text{Li} + \nu_e \\ \text{(V)} \quad {}^7\text{Li} + {}^1\text{H} \xrightarrow{\text{decay}} {}^4\text{He} + {}^4\text{He} \end{array}$$

PPIII-Branch 0.02 % in Sun.

$$\begin{array}{l} \text{(IV)} \quad {}^7\text{Be} + {}^1\text{H} \rightarrow {}^8\text{B} + \gamma \\ \text{(V)} \quad {}^8\text{B} \xrightarrow{\text{decay}} {}^8\text{Be} + e^+ + \nu_e \\ \text{(VI)} \quad {}^8\text{Be} \xrightarrow{\text{decay}} {}^4\text{He} + {}^4\text{He} \end{array}$$

CNO-Cycle vs PP-Chain (Carbon,Nitrogen,Oxygen)

PP-Chain	CNO-Cycle
$10^7 \text{ K} < T < 10^{17} \text{ K}$	CNO: $T > 10^{17} \text{ K}$
$M \lesssim 1.3 M_\odot$	$M \gtrsim 1.3 M_\odot$
$\epsilon_{\text{PP}} \propto \rho X^2 T_6^4$	$\epsilon_{\text{CNO}} \propto \rho X Z \text{CNO} T_6^{20}$
Dominant in Sun	Dominant in massive stars



Massive stars $\implies$ CNO cycle, more temperature sensitive $\implies$ convective core in massive stars.
Small stars $\implies$ PP-Chain $\implies$ radiative core, convective envelope.

3.8 Star Phenomena

Binary Stars Two stars orbiting common center of mass. If material extends beyond Roche lobe, mass transfer occurs between stars.

Stellar Rotation Plasma rotating around star's axis, causing northpole to rotate slower than equator. Magnetic field lines are twisted by differential rotation and suppress

convection locally $\implies$ sunspots

Stellar Winds Mass loss due to radiation pressure on outer layers. Typical mass loss rate: $10^{-14} M_\odot/\text{yr}$ for Sun-like stars, up to $10^{-4} M_\odot/\text{yr}$ for massive stars.

Planetary Nebula Ejection of outer layers of low to intermediate mass stars during late stages of evolution. Leaves behind a white dwarf.

Supernovae Explosive death of massive stars. Types: Synthesis of heavier elements than Fe.

Type	Mechanism	Note
Type II	Core collapse massive stars	Neutron star/Black hole
Type Ia	White dwarf binary system	Chandrasekhar limit ($1.44 M_\odot$)

4 Milky Way Galaxy

4.1 Structure

- Galactic Disk:** Flat, rotating disk with spiral arms, containing gas, dust, and stars. Supported by **rotation**.
- Galactic Bulge:** Central, spherical region with a supermassive black hole (Sagittarius A*). Supported by **random motions**.
- Galactic Halo:** Spherical region surrounding the disk and bulge. Supported by **random motions**.

Disk Rotation Angular velocity $\omega(R) = \frac{v_{\text{rot}}(R)}{R}$ is variable with radius R . Density waves cause spiral arms.

4.2 Extinction

Interstellar matter absorbs star light, causing **dimming** and **reddening**.

modified distance modulus:

$$m - M = 5 \log(d/10\text{pc}) + A_\lambda \quad \text{with } A_\lambda \text{ extinction in magnitudes.}$$

Extinction For V magnitude: $A_V \approx 1.5 \left(\frac{d}{1\text{kpc}}\right)$ mag.

Reddening

$$(B-V)_{\text{obs}} = (B - V) - E_{B-V} \quad \text{with } E_{B-V} \approx 0.5 \left(\frac{d}{1\text{kpc}}\right) \text{ mag.}$$

$\frac{A_\lambda}{E_{B-V}}$ is independent of distance $\implies$ extinction measure can probe **composition of interstellar matter**.

4.3 Spectra Lines: (u upper, l lower, energy level)

Line Profile $\Phi(\Delta\nu)$

Spectral lines are broadened due to **doppler broadening, natural/lifetime broadening and collisional broadening**. $n_u = \frac{g_u}{g_l} \exp\{(\hbar\nu_0/kT)\}$
 $\int \Phi(\Delta\nu) d\nu = 1$

Einstein Coefficients (n_x number of x)			
	Process	Rate	Note
u $\rightarrow$ l	Spontaneous emission	$n_u = A_{ul} n_u$	indep. of rad. field
l $\rightarrow$ u	Absorption	$n_l = B_{lu} n_l n_\nu$	$\propto$ energy density n_ν
u $\rightarrow$ l	Stimulated emission	$n_u = B_{ul} n_u n_\nu$	$\propto$ energy density n_ν

Radiative Transfer The 2-Level model can be used in a radiative transfer equation to describe the change in intensity I_ν through the ISM.

Emissivity	$j_\nu = \frac{\hbar\nu_0 n_u A_{ul}}{4\pi} \Phi(\Delta\nu)$
Absorption	$\alpha_\nu = \frac{\hbar\nu_0}{4\pi} (n_l B_{lu} - n_u B_{ul}) \Phi(\Delta\nu)$
Source	$S_\nu = \frac{j_\nu}{\alpha_\nu} = \frac{n_u A_{ul}}{n_l B_{lu} - n_u B_{ul}}$
$S_\nu = B_\nu(T) \quad g_l B_{lu} = g_u B_{ul} \quad A_{ul} = \frac{8\pi\hbar\nu_0^3}{c^3} B_{ul}$	

21cm Line Hyperfine transition of HI from parallel to antiparallel ground spin state. ($\hbar\nu \ll kT$)

Scenario	Condition	Note
Emission	$\alpha_\nu = 0$	$I = \frac{3}{16\pi} \hbar\nu_0 A_{ul} \int n_H ds$
Absorption	$j_\nu = 0$	$\tau_\nu = \frac{3}{32\pi} \frac{\hbar c^2 A_{ul}}{k_B \nu_0} \Phi(\Delta\nu) \int \frac{n_H}{T} ds$

Probe n_H and T in the ISM. $\Leftarrow$ **Absorption & Emission** spectral $A_{ul}^{-1} \approx 10^7 \text{ yrs}$, $\lambda = 21.1 \text{ cm}$, $\nu_0 = 1.42 \text{ GHz}$.

Visible Spectrum Optical lines (e.g. H α , H β) are used to probe HII in ISM. ($\hbar\nu \sim 1\text{eV} \gg kT$)

$$\left. \tau_\nu = \frac{\hbar\nu_0}{c} B_{lu} \Phi(\Delta\nu) N \right| \quad \text{column density: } N = \int n ds \quad \left[\text{cm}^{-2} \right]$$

Equivalent Width $W_{\lambda_0} = \int d\lambda \frac{I_c - I_\nu}{I_c} \simeq \frac{\lambda_0^2}{\int d\nu} \frac{I_c - I_\nu}{I_c}$
 $I_c \equiv$ **continuum background intensity** $\implies W_{\lambda_0} \equiv$ **normalized area of spectral line**.

$$\tau_\nu \ll 1: W_{\lambda_0} = \frac{\lambda_0^2 \hbar\nu_0}{c^2} B_{lu} N \implies \left[W_{\lambda_0} / \lambda_0 \simeq \frac{\hbar}{c} B_{lu} N \right]$$

4.4 Interstellar Medium

Matter between stars within a galaxy.

HI Regions	WIM	Coronal Gas
$T \sim 80 \text{ K}$	$T \sim 8000 \text{ K}$	$T \sim 10^6 \text{ K}$
$n \sim 10^6 - 8 \text{ m}^{-3}$	$n \sim 10^{5-6} \text{ m}^{-3}$	$n \sim 10^3 \text{ m}^{-3}$
$n_{\text{metal}} \ll n_H$	like hot a HI	$\ll$ ISM t_{tot} mass
$f_V \sim 5\%$	$f_V \sim 40\%$	emits $\gamma \sim \text{keV}$
		$f_V \sim 50\%$

HII Regions (ionized)	Molecular Clouds
$T \sim 6000 \text{ K}$	$T \sim 10 - 30 \text{ K}$
Strömgren radius	$n \sim 10^9 \text{ m}^{-3}$
$R_S = \left(\frac{3N_\gamma}{4\pi\alpha n_p n_e}\right)^{1/3}$	H ₂ , CO, OH
UV $\sim 912 \text{ \AA}$	Radio, vis, UV
	$f_V \sim 1\%, f_M \sim 40\%$

5 Stellar Dynamics

System of self-gravitating particles: Only force acting on particle i is gravitational force from other particles.

5.1 Virial Theorem

Consider a bound, self-gravitating system of many particles with positions $\vec{x}_i$, masses m_i and velocities $\vec{v}_i$. The total kinetic energy is $K = \sum_i \frac{1}{2} m_i v_i^2$ and the total gravitational potential energy is $V = -\sum_{i < j} \frac{G m_i m_j}{|\vec{x}_i - \vec{x}_j|}$. By relating kinetic and potential energy through the equations of motion and performing a time average, one obtains the virial theorem, which connects the long-term averaged energies of the system:

$$2\overline{K} + \overline{V} = 0$$

With the total energy $E = K + V$, the virial theorem gives:

$$\overline{E} = -\overline{K} = \frac{1}{2} \overline{V}$$

When does the virial theorem apply? Virial theorem applies when system is **virialized** i.e. in **steady state!**

Self-gravitating system Consider a system of characteristic size (R) consisting of N stars, each of mass m . The total kinetic energy is $\frac{1}{2} m N \langle v^2 \rangle$

where $\langle v^2 \rangle$ denotes the mean square stellar velocity (velocity dispersion).

The **total gravitational potential energy** is

$$\overline{V} \approx -N_{\text{pairs}} \frac{Gm^2}{\langle R \rangle} = -\frac{N(N-1)}{2} \frac{Gm^2}{\langle R \rangle} \simeq -\frac{G(mN)^2}{2\langle R \rangle},$$

where N_{pairs} is the number of particle pairs and N is assumed to be large. Using the virial theorem and the total mass $M = mN$, one obtains

$$\langle v^2 \rangle \simeq \frac{GM}{2\langle R \rangle}$$

Specific heat (self-gravitating system) By equipartition, the mean kinetic energy per degree of freedom is $\frac{1}{2} k_B T$. A system of N particles has $3N$ translational degrees of freedom, hence

$$\overline{K} = \frac{3}{2} N k_B T.$$

(constant volume) specific heat: $c_V \equiv \frac{d\overline{E}}{dT}$

For a virialized self gravitating system $\overline{E} = -\overline{K}$ so that:

$$c_V = -\frac{d\overline{K}}{dT} = -\frac{3}{2} N k_B < 0$$

Therefore, self-gravitating systems have **negative specific heat**: if the total energy decreases (becomes more negative), the system contracts ($\langle R \rangle$ decreases) and heats up ($\langle v^2 \rangle$ and T increase). Conversely, adding energy causes the system to expand and cool.

5.2 Collisional Relaxation

In ordinary gases, TE is reached through frequent short-range particle collisions. But in stellar systems, stars interact via long-range gravity and direct physical collisions are extremely rare. Relaxation towards TE therefore proceeds through cumulative gravitational encounters between stars called *two-body (collisional) relaxation*.

Two-body encounters Consider two stars of mass m interacting gravitationally with relative velocity v and impact parameter b . The typical momentum of a star is $p \approx mv$

During the encounter, the gravitational force F is acting over a characteristic time Δt :

$$F \approx \frac{Gm^2}{b^2}, \quad \Delta t \approx \frac{b}{v}$$

The resulting change in momentum is therefore:

$$\Delta p \approx F \Delta t = \frac{Gm^2}{bv}$$

A close encounter (collision) is defined by $\Delta p \gtrsim p$, implying a characteristic collision distance:

$$\Delta p \gtrsim p \implies b \lesssim b_{\text{col}} \equiv \frac{Gm}{v^2}$$

Gravitational cross section The effective cross section for such encounters is given by

$$\sigma \approx \pi b_{\text{col}}^2 = \pi \frac{G^2 m^2}{v^4}$$

This gravitational cross section is much larger than the geometric cross section of stars, but still leads to **long relaxation times in systems with many particles**.

Relaxation time Let n be the number density of stars. The encounter rate for a given star is $\Gamma \approx n \sigma v$, and the corresponding relaxation time is

$$T_{\text{rel}} \equiv \Gamma^{-1} \approx \frac{1}{n \sigma v} = \frac{v^3}{\pi G^2 n m^2}$$

This timescale characterizes the time required for a star to undergo sufficient encounters to significantly change its velocity.

System	v [km s ⁻¹]	n [pc ⁻³]	T_{rel} [yr]
Galaxy	100	0.1	10 ¹⁷
Open cluster	0.5	1	10 ⁹
Globular cluster	10	10 ³	10 ¹⁰

Crossing time and scaling with N Another important timescale is the crossing time,

$$T_{\text{cross}} \sim \frac{R}{v},$$

where R is the characteristic size of the system. For a virialized system $\langle v^2 \rangle$ is known and Using $n \sim N/R^3$, one finds the scaling

$$\frac{T_{\text{rel}}}{T_{\text{cross}}} \sim N$$

For large- N systems (galaxies), T_{rel} exceeds the age of the Universe, making them effectively collisionless, while for small- N systems (globular clusters) two-body relaxation is dynamically important. Hence, galactic evolution cannot be described by collisional relaxation and is governed by other processes. But even the more efficient relaxation processes do not bring galaxies into full TE!

Violent relaxation During rapid gravitational collapse, the gravitational potential varies strongly in time. This leads to efficient mixing in phase space, a process known as *violent relaxation* (Lynden-Bell). It's characteristic timescale is (T_{VR} a few T_{cross}), MUCH shorter than the two-body relaxation time.

Dynamical friction A massive object moving through a stellar system will attract other particles along the way. The particle density will be enhanced in its wake, resulting in a net gravitational force backwards, a drag force known

as *dynamical friction* (Chandrasekhar). This process (acting on short time-scales) causes massive objects to lose energy and sink toward the center of the system.

5.3 Thermodynamic equilibrium and self-gravity

Boltzmann distribution (canonical ensemble):

$$P(E) \propto e^{-\beta E}, \quad \beta = \frac{1}{k_B T}$$

Partition function: $Z = \sum_i e^{-\beta E_i}$

Mean Energy: $\langle E \rangle = \frac{1}{Z} \sum_i E_i e^{-\beta E_i}$

$$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta}$$

$$\langle E^2 \rangle = \frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2}$$

$$(\Delta E)^2 = \langle E^2 \rangle - \langle E \rangle^2 = \frac{\partial^2 \ln Z}{\partial \beta^2}$$

$$c_V = \frac{\partial \langle E \rangle}{\partial T} = -k_B \beta^2 \frac{\partial \langle E \rangle}{\partial \beta} = k_B \beta^2 (\Delta E)^2$$

$$c_V = k_B \beta^2 (\Delta E)^2 > 0$$

Because $c_V > 0$ in the canonical ensemble, a system in thermal contact with a heat bath cannot have negative specific heat. Therefore, self-gravitating systems cannot relax to canonical TE; this incompatibility is due to the attractive, long-range nature of gravity.

5.4 Boltzmann Equation

Distribution function $f(\vec{x}, \vec{v}, t) d^3x d^3v$ = number of particles in phase space (6-dimensional) volume $d^3x d^3v$ about $(\vec{x}, \vec{v})$ at time t .

Phase-space flow (Liouville's Thrm.) $f(\vec{x}(t), \vec{v}(t), t)$ is constant along trajectories, (phase-space flow is incompressible). Along particle trajectories in phase space, the distribution function is conserved so its total time derivative vanishes,

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_i \frac{\partial f}{\partial x_i} \frac{dx_i}{dt} + \sum_i \frac{\partial f}{\partial v_i} \frac{dv_i}{dt} = 0,$$

which is the collisionless Boltzmann equation. In general, collisions add a source term, where $C[f]$ is the collision operator;

$$\frac{df}{dt} = C[f],$$

Self-gravitating systems For typical self-gravitating systems collisions are often inefficient, so you can neglect $C[f]$. The only force $\vec{F}$ acting on the particles is the gravitational force. Thus the collisionless Boltzmann equation becomes:

$$\frac{\partial f}{\partial t} + \sum_i \frac{\partial f}{\partial x_i} v_i - \sum_i \frac{\partial f}{\partial v_i} \frac{\partial \Phi}{\partial x_i} = 0$$

$$\nabla^2 \Phi = 4\pi G \rho$$

$$\rho(\vec{x}, t) = m \int f(\vec{x}, \vec{v}, t) d^3v$$

The three equations above form a coupled system of integro-differential equations for f , ρ , and Φ . They implicitly assume a smooth density and potential, whereas the true gravitational potential is

$$\Phi_{\text{actual}}(\vec{x}, t) = -\sum_{\alpha} \frac{Gm}{|\vec{x} - \vec{x}_{\alpha}(t)|} \neq \Phi_{\text{smooth}}(\vec{x}, t)$$

The difference $\Phi_{\text{actual}} - \Phi_{\text{smooth}}$ gives rise to a collision term when relevant.

Jeans theorem (steady-state solutions) In steady state the distribution function is **time-independent**, $f(\vec{x}, \vec{v}, t) = f(\vec{x}, \vec{v})$. Collisionless Steady-state solutions can be constructed from integrals of motion $C_k(\vec{x}, \vec{v})$ satisfying:

$$\frac{dC_k}{dt} = 0 \quad \text{along trajectories.}$$

If f depends on phase space only through these constants, $\Rightarrow f = f(C_k(\vec{x}, \vec{v}))$,

$$\text{then} \quad \frac{df}{dt} = \sum_k \frac{\partial f}{\partial C_k} \frac{dC_k}{dt} = 0,$$

so the collisionless Boltzmann equation is automatically satisfied (since $dC_k/dt = 0$). Examples: energy, angular momentum.

Widely used steady-state solutions For a time-independent potential $\Phi(\vec{x})$ (and neglecting collisions), the single-particle energy

$$E(\vec{x}, \vec{v}) = \frac{1}{2} m v^2 + m \Phi(\vec{x})$$

is a constant of motion. By Jeans theorem, any distribution function that depends on phase space only through E provides a steady-state solution of the collisionless Boltzmann equation, i.e.

$$f(\vec{x}, \vec{v}) = f(E)$$

Singular Isothermal Sphere (SIS) A widely used steady-state, spherically symmetric isothermal solution has:

$$\frac{1}{l^2} \frac{d}{dl} \left(\frac{l^2}{\xi} \frac{d\xi}{dl} \right) = -9\xi \xrightarrow{\xi(l) = \frac{2}{9l^2}} \rho(r) = \frac{2}{9} \rho_c \left(\frac{r_0}{r} \right)^2$$

It is *singular* at $r=0$ ($\rho \rightarrow \infty$) and has divergent total mass since

$$M(r) = 4\pi \int_0^r \rho(r') r'^2 dr' \propto r \quad (r \rightarrow \infty),$$

so it must be truncated in practice. For the SIS the velocity dispersion is constant, $\langle v^2 \rangle = 3\sigma^2$.

In general first compute $M(r)$ then equate $F_{\text{ZP}} = m v_c^2 / r$ and $F_G = GmM(r)/r^2$, and solve for $v_c(r) = \sqrt{GM(r)/r}$. ($v = \omega r$). For this specific case $M(r) \propto r$, circular velocity is approx. constant (*flat rotation curve*):

$$v_c(r) = \sqrt{\frac{GM(r)}{r}} \sim \text{constant}$$

King model (truncated SIS) A more realistic collisionless steady-state solution is the *King model*,

$$f(E) = \begin{cases} f_0 \left[\exp\left(-\frac{E + m\Phi_0}{m\sigma^2}\right) - 1 \right], & E \leq m\Phi_0, \\ 0, & E > m\Phi_0, \end{cases}$$

with parameters f_0, σ, Φ_0 . It must be integrated numerically and yields a truncated density profile with $\rho(r) = 0$ for $r > r_t$, hence finite total mass. Provides good fits for bright ellipticals.

6 Fluid & Plasma Astrophysics

6.1 Fluid Mechanics

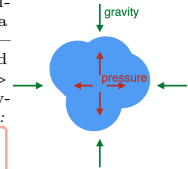
Describes dynamics of neutral gas using 4 equations.

Continuity (Mass)	$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$
Euler (Momentum)	$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} = -\frac{1}{\rho} \nabla P + \vec{F}$
State (Adiabatic)	$\frac{d}{dt} \left(\frac{P}{\rho^\gamma} \right) = 0 \quad (\gamma = 5/3)$
Poisson (Gravity)	$\vec{F} = -\vec{\nabla} \Phi \quad \nabla^2 \Phi = 4\pi G \rho$

6.2 Gravitational Instability

Initial **mass inhomogeneity** collapses to form a star or a galaxy. $\Rightarrow$ **Jeans instability** — Competition between **gravity** and **pressure** $|k| > k_J$: stable (Press. > Grav.) Oscillation | else exp. growth/decay | *Perturbation treatment*:

$$\rho_1 \ll \rho_0, P_1 \ll P_0, |\vec{v}_1| \ll 1 \\ \vec{v}_0 = 0, \phi_1 \ll \phi_0$$



0th Order (Jeans Swindle)

Static homogeneous fluid $\Rightarrow \rho_0 = \text{const}, P_0 = \text{const}$. **Euler:** $\vec{\nabla} P_0 = -\rho_0 \vec{\nabla} \Phi_0 \Rightarrow \vec{\nabla} \Phi_0 = 0 \Rightarrow \Phi_0 = \text{const}$. **Poisson:** $\nabla^2 \Phi_0 = 4\pi G \rho_0 \Rightarrow \rho_0 = 0$ for $\Phi_0 = \text{const} \Rightarrow \text{!}$

$\rho_0 = 0$ to satisfy both equations $\Rightarrow$ contradicts homogeneous fluid $\Rightarrow$ **Jeans Swindle**: assume $\rho_0 \neq 0$ and proceed with perturbation analysis.

1st Order Perturbation

Continuity	$\rho = \rho_0 + \rho_1 \Rightarrow \frac{\partial \rho_1}{\partial t} + \rho_0 \nabla \cdot \vec{v}_1 = 0$
Euler	$\vec{v} = \vec{v}_0 + \vec{v}_1 \Rightarrow \frac{\partial \vec{v}_1}{\partial t} = -\frac{c_s^2}{\rho_0} \vec{\nabla} \rho_1 - \vec{\nabla} \Phi_1$
State	$P = P_0 + P_1 \Rightarrow P_1 = c_s^2 \rho_1$ with $c_s^2 = \gamma P_0 / \rho_0$
Poisson	$\Phi = \Phi_0 + \Phi_1 \Rightarrow \nabla^2 \Phi_1 = 4\pi G \rho_1$

Fourier Decompositions: $\rho_1(\vec{x}, t) = \sum_{\vec{k}, \omega} \hat{\rho}_1(\vec{k}, \omega) e^{i(\vec{k} \cdot \vec{x} - \omega t)}$

Continuity	$-i\omega \hat{\rho}_1 + i\rho_0 \vec{k} \cdot \vec{v}_1 = 0$	Fourier Space: $\frac{\partial}{\partial t} \rightarrow -i\omega$ $\frac{\partial}{\partial \vec{x}} \rightarrow i\vec{k}$
Euler	$-i\omega \vec{v}_1 = -\frac{c_s^2}{\rho_0} i\vec{k} \hat{\rho}_1 - i\vec{k} \hat{\Phi}_1$	
Poisson	$-k^2 \hat{\Phi}_1 = 4\pi G \hat{\rho}_1$	

1. substitute $\hat{\Phi}_1$ from Poisson into Euler and rearrange for $\vec{v}_1 \rightarrow 2. \left[-\omega^2 + k^2 c_s^2 - 4\pi G \rho_0 \right] \hat{\rho}_1 = 0 \rightarrow 3. [\dots] = 0$ and rearrange for $\omega^2 = c_s^2 (k^2 - \frac{4\pi G \rho_0}{c_s^2}) = c_s^2 (k^2 - k_J^2)$

The critical wavenumber k_J decides whether perturbations grow or die over time. $\Rightarrow$ **initial mass & size**

Jeans Criterion

Jeans Length	$\lambda_J = \frac{2\pi}{k_J} = \sqrt{\frac{\pi c_s^2}{G \rho_0}}$	 $\lambda = 2\pi/k$
Jeans Mass	$M_J = \frac{4}{3} \pi \lambda_J^3 \rho_0 = \frac{4}{3} \pi \frac{5}{2} \frac{c_s^3}{G^{3/2} \rho_0^{1/2}}$	

HI Cloud: $T \simeq 100$ K, $n \simeq 10^6$ m⁻³ $\Rightarrow M_J \simeq 10^5 M_\odot$
Mol. Cloud: $T \simeq 10$ K, $n \simeq 10^9$ m⁻³ $\Rightarrow M_J \simeq 400 M_\odot$

Scenario	Wavenumber	Length	Mass
Stable	$k > k_J$	$\lambda < \lambda_J$	$M < M_J$
Unstable	$k < k_J$	$\lambda > \lambda_J$	$M > M_J$

6.3 Plasma Effects

Bremsstrahlung In a fully ionized, equilibrated plasma, free electrons are accelerated in the Coulomb field of ions, emitting thermal **Bremsstrahlung** $\epsilon_\nu \propto \rho_{\text{gas}}^2$.

$$\epsilon_\nu \simeq 6.8 \cdot 10^{-51} \frac{n_e n_i Z^2}{\sqrt{T}} e^{-\frac{h\nu}{kT}} g(\nu, T) \quad \left[\frac{\text{W}}{\text{m}^3 \text{Hz}} \right]$$

HII regions emit in **radio band**. Hot gas in galaxies and ionized intracluster medium (ICM) emits in **X-ray band** ($T \sim 10^7$ - 10^8 K).

Synchrotron Radiation

In hot plasma with **magnetic fields**, free electrons spiral along field lines, the **acceleration** causes emission of **synchrotron radiation**. Supernova remnants, Radio galaxies, Quasars, Hot ISM
Powerlaw energy distribution:
 $N(E) dE = N_0 E^{-p} dE$

Ionization Fraction Mole fraction x of ionized atoms in a thermal plasma. Ionization potential χ .

$$\frac{x}{1-x} = \frac{(2\pi m_e)^{3/2} (k_B T)^{5/2}}{h^3 P} e^{-\frac{\chi}{kT}}$$

7 Extragalactic Astrophysics

7.1 Normal Galaxies

Normal Galaxy: Galaxy with emission spectrum looking like a composite of light from many stars.

Galaxy Morphology (Hubble classification): Elliptical galaxies: E0-E7 with increasing apparent ellipticity (E0 nearly round, E7 strongly elongated). Lenticular galaxies (S0): a disk and bulge but no visible spiral arms. Spiral galaxies: classified as Sa, Sb, Sc (and Spiral barred SBa, SBb, SBc), where the sequence corresponds to decreasing bulge size and increasingly open spiral arms (Sa tight arms, Sc open arms).

Luminosity Function The luminosity function $\Phi(L) dL$ gives the number of galaxies per unit volume with luminosities in $[L, L + dL]$. For ellipticals it is well described by the *Schechter function*

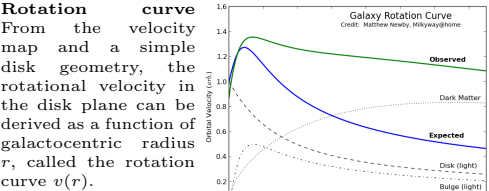
$$\Phi(L) = N_0 \left(\frac{L}{L_*}\right)^\alpha \exp\left(-\frac{L}{L_*}\right) \frac{1}{L_*},$$

with typical parameters $N_0 \simeq 1.2 \times 10^{-2} h^3 \text{Mpc}^{-3}$, $\alpha \simeq -1.25$, $L_* \simeq 10^{10} h^{-2} L_\odot$. Spirals exhibit a wider range of luminosities.

Physical Characteristics of Galaxies

Property	Ellipticals (E)	Spirals (S)
ISM / gas	very little ISM (gas-poor)	lots of ISM in disk (gas-rich)
Stellar population	Population II-like (old)	disk: young + old; bulge: older
Colour	redder (no young blue stars)	disk bluer; bulge redder
Dynamical support	random motions (pressure, high σ)	Halo & bulge random motion; disk rotational support
Star formation	low / quenched	active in disk/arms
Magnetic fields	not prominent	B fields + CR $\rightarrow$ synchrotron
Scaling law	Faber–Jackson: $\sigma \propto L^{0.25}$	Tully–Fisher: $v_c \propto L^{0.22}$

Velocity mapping (Doppler) For a rotating galactic disk inclined with respect to the line of sight, the side whose orbital motion has a component toward the observer is Doppler blue-shifted, while the opposite side is red-shifted. By measuring the Doppler shift of spectral lines (e.g. 21 cm line) at many positions (x, y) across the galaxy, one obtains the velocity field (velocity map) of the galaxy.



For circular orbits, the circular velocity is $v_c(r) = \sqrt{GM(r)/r}$, (where $M(r)$ is the mass enclosed within radius r) as introduced in the SIS model.

Expected vs. observed behaviour (spiral galaxies)
If the mass density is approximately constant ($\rho \simeq \text{const}$), then inside the galaxy $M(r) \propto r^3$ and $v_c(r) \propto r$. Outside the galaxy ($\rho \simeq 0$), one would expect $M(r) = \text{const}$ and a Keplerian decline $v_c \propto r^{-1/2}$.
Observed spiral galaxies, however, show flat rotation curves ($v_c \simeq \text{const}$) well beyond the stellar disk, implying $M(r) \propto r$ and therefore $\rho(r) \propto r^{-2}$ (like the SIS model), consistent with the presence of an extended, non-luminous dark matter halo.

Ellipticals (Faber–Jackson): the stellar velocity dispersion σ correlates with luminosity,
$$\sigma \approx 220 \left(\frac{L}{L_*}\right)^{0.25} \text{ km/s}$$

Spirals (Tully–Fisher): the circular velocity v_c in the flat part of the rotation curve correlates with luminosity,
$$v_{c,\text{flat}} \approx 220 \left(\frac{L}{L_*}\right)^{0.22} \text{ km/s}$$

7.2 Expansion of the Universe

Doppler effect (waves) A light source moving away ($v > 0$, positive) is observed at lower frequency (longer wavelength), while a source moving closer ($v < 0$) is observed at higher frequency (shorter wavelength). $\nu_{\text{obs}}, \nu_{\text{em}}$ are the observed and emitted frequencies. By using ($\nu = c/\lambda$) we can get the Redshift (of light):

$$\frac{\nu_{\text{em}}}{\nu_{\text{obs}}} = \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} \simeq \frac{c+v}{c} = 1 + \frac{v}{c} = 1+z$$

Redshift: $z = \frac{v}{c}$

Hubble–Lemaître law More distant galaxies have larger redshifts, i.e. they recede faster. For nearby galaxies the recession velocity scales linearly with distance, $v \simeq H_0 l$ (l small), where H_0 is the Hubble constant (often called Hubble *parameter* since it evolves with cosmic time). In a uniformly expanding universe, any observer sees the same relation $v \propto l$ (no special center).

Peculiar velocities Galaxies follow the Hubble flow, but they also have local motions ("peculiar velocities"). A simple model is

$$\vec{v} \simeq H_0 \vec{l} + \delta \vec{v}$$

where $H_0 l$ is the expansion term and δv is the peculiar velocity (typically $\delta v \sim 500 \text{ km s}^{-1}$). The Hubble constant H_0 is determined from galaxy recession velocities v (redshift) and distances l (Cepheids, Type Ia supernovae, Tully–Fisher/Faber–Jackson, gravitational lensing). For small l , peculiar velocities matter. For large l , distance uncertainties dominate.

Hubble constant H_0 and time
$$H_0 = 100 h \text{ km s}^{-1} \text{ Mpc}^{-1},$$

with the dimensionless parameter $h \simeq 0.7$. For small redshifts ($z \ll 1$),
$$l = \frac{v}{H_0} = \frac{c}{H_0} z \simeq 3000 z h^{-1} \text{ Mpc}.$$

Since H_0 has units of inverse time, one defines the Hubble time which is of the order of the age of the Universe ($\sim 14 \text{ Gyr}$):
$$H_0^{-1} \simeq 9.8 \times 10^9 h^{-1} \text{ yr},$$

7.3 Active Galaxies

The luminosity of active galaxies is dominated by extremely bright active galactic nuclei (AGN) whose emission spectra differ from those of normal galaxies. There are several types of AGNs:

Seyfert galaxies: Spiral galaxies with unusually bright nuclei (strong emission). Type 1: broad emission lines; Type 2: narrow emission lines.

Radio galaxies: Galaxies with strong radio emission from 2 large lobes outside the galaxy. The lobes are produced by jets from the central nucleus interacting with the intergalactic medium (gas). Radio emission is synchrotron radiation (power-law spectrum), indicating magnetic fields. Emission lines can be broad or narrow.

Quasars (QSOs): Extremely luminous, compact AGNs, often at high redshift. Appear star-like in optical images. Luminosities up to $L \sim 10^{39} \text{ W}$, 100 times brighter than normal galaxies. Rapid emission variability ($\Delta t \sim \text{days}$) implies a very small emitting region ($\Delta x \lesssim c \Delta t$). Optical spectra resemble Seyfert 1 galaxies (broad lines), Seyferts being milder. Radio-quiet quasars (Quasi-Stellar Objects, QSO) emit in visible / UV instead of radio.

Central engine of AGN AGN are powered by accretion onto a central supermassive black hole, where Thomson scattering (for a neutral plasma) dominates:

$$\chi = \frac{n_e \sigma T}{\rho} = \frac{\sigma T}{m_H}$$

The luminosity is limited by the Eddington luminosity, obtained from the balance between gravity and radiation pressure,

$$L_{\text{Edd}} = \frac{4\pi c G M m_H}{\sigma T} \simeq 1.3 \times 10^{31} \left(\frac{M}{M_\odot}\right) \text{ W}.$$

Observed quasar luminosities ($L \sim 10^{39} \text{ W}$) imply black hole masses $M \sim 10^8 M_\odot$. The characteristic size is set by the Schwarzschild radius,

$$r_s = \frac{2GM}{c^2} \simeq 3.0 \times 10^3 \left(\frac{M}{M_\odot}\right) \text{ m}$$

If the black hole emits blackbody radiation at temperature T , the luminosity at the Schwarzschild radius is given by the Stefan–Boltzmann law, $L = 4\pi r_s^2 \sigma T^4$. Setting $L = L_{\text{Edd}}$ yields the effective black hole temperature,

$$T = \left(\frac{c^5 m_H}{4 \sigma \sigma T G M}\right)^{1/4} \simeq 3.7 \times 10^5 \left(\frac{10^8 M_\odot}{M}\right)^{1/4} \text{ K}$$

Mass accretion onto black holes: Accretion converts rest mass into radiation with efficiency η ,

$$\frac{dM}{dt} \simeq \frac{L}{\eta c^2} \simeq 1.5 \left(\frac{0.1}{\eta}\right) \left(\frac{L}{10^{39} \text{ W}}\right) M_\odot \text{ yr}^{-1}.$$

Typical $\eta \sim 0.1$ implies $\dot{M} \sim 1 M_\odot \text{ yr}^{-1}$ is sufficient to power quasars.

AGN Unification scheme: The many different AGN types are the same kind of object, but viewed from different angles, times (active or inactive phase).

7.4 Clusters of galaxies and large scale distribution of galaxies

Local Group: less dense, irregular shape/distribution, smaller mass, less galaxies (~ 35).

Galaxy Clusters: very dense, regular shape, bigger mass ($\sim 10^{15} M_\odot$), many galaxies ($\gtrsim 100$), largest bound structures in the universe ($\sim 1 \text{ Mpc}$), luminosity $\sim 10^{13} L_\odot$.

cD Galaxies: Large elliptical galaxies at the center of galaxy clusters, resulting of mergers of other galaxies trough dynamical friction ("galactic cannibalism").

Gas cooling, cooling flows and origin of the ICM. For a fully ionized hydrogen plasma in galaxy clusters the particle number is $N \simeq 2n_e (4\pi/3) R^3$, giving a thermal energy $E \simeq N k_B T$ (up to factors of order unity). If X-ray emission is dominated by thermal Bremsstrahlung, the cooling time is $t_{\text{cool}} \simeq \frac{E}{L_X}$, $\epsilon_{\text{ff}} \propto n_e n_p \propto \rho_{\text{gas}}^2$.

Typical cluster values ($n_e \sim 10^3 \text{ m}^{-3}$, $R \sim 1 \text{ Mpc}$, $T \sim 10^8 \text{ K}$) yield $t_{\text{cool}} \sim 10^{11} \text{ yr} \gg t_U$, so the gas remains hot on cosmological timescales, although cooling is more efficient in dense cores, where inward "cooling flows" are predicted but weakly observed, implying additional heating. The intracluster medium originates from gas trapped during cluster formation and heated by conversion of gravitational potential energy, $\eta G_{\text{gc}} m_p / R_{\text{gc}} \sim k_B T$, giving $T \sim 10^8 \text{ K}$ for massive clusters, it is enriched by ISM material via galaxy mergers and ram-pressure stripping.

8 Cosmology

8.1 General Relativity

Gravity is not a force, but a **curvature of a 4-dimensional spacetime manifold** caused by mass and energy.

Principle of Equivalence Locally, gravitational acceleration is indistinguishable from acceleration caused by other forces. $\Rightarrow$ Free-falling observers experience no gravity. $\vec{F} = m \nabla \Phi$, $\vec{a} = \nabla \Phi$ $\Rightarrow$ Acceleration is independent of mass of particle. $\Rightarrow$ Gravity is property of spacetime.

Cosmological Principles
Universe is homogeneous and isotropic on large scales ($\gtrsim 100 h^{-1} \text{ Mpc}$).
 $\Rightarrow$ No special locations or directions in the universe.
 $\Rightarrow$ homogeneous & isotropic spaces have uniform curvature.

Only 3 possible geometries for **homogeneous & isotropic** space $\Rightarrow$ **Comoving coordinates** to describe evolving universe $\Rightarrow$ unify geometries with **curvature constant k** $\Rightarrow$ combine time component with spatial metric $\Rightarrow$ **FLRW metric**.

Curvature *

Curvature (Δ)	Geometry	Example
Positive ($>180^\circ$)	Spherical	Surface of a sphere
Negative ($<180^\circ$)	Hyperbolic	Saddle surface
Zero ($=180^\circ$)	Flat	Euclidean plane

Metric *

Geometry	Metric
Euclidean	$ds^2 = dx^2 + dy^2 + dz^2 = a^2(dx^2 + \chi^2 d\Omega^2)$
Spherical	$ds^2 = a^2(dx^2 + \sin^2 \chi d\Omega^2)$
Hyperbolic	$ds^2 = a^2(dx^2 + \sinh^2 \chi d\Omega^2)$

where $\chi = \frac{r}{a}$, scale factor **a** , $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$ solid angle element.

Unified Spatial Metrics *

$$ds_{\text{space}}^2 = a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right) = \begin{cases} \sin \chi & k = +1 \\ \chi & k = 0 \\ \sinh \chi & k = -1 \end{cases}$$

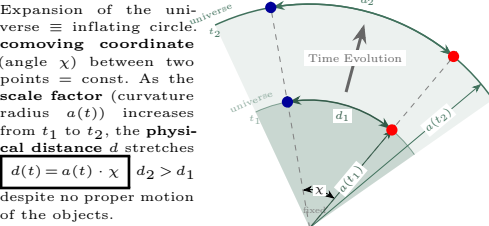
with k curvature constant $\Rightarrow$ compare 3 geometries in single metric.

Friedmann–Lemaître–Robertson–Walker (FLRW) Metric *

The scaled metric using the comoving coordinates a, χ and $d\Omega$ is handy to describe the evolving universe.

Add time component to spatial metric:

$$ds^2 = -c^2 dt^2 + a(t)^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right)$$



8.2 Friedmann Equations

Combine FLRW with EFE to describe dynamics of expanding universe.

Hubble Parameter & Redshift
 $\mathbf{v} = \frac{d}{dt}(a\chi) = \dot{a}\chi = \frac{\dot{a}}{a}(a\chi) = H d \Rightarrow H(t) = \frac{\dot{a}(t)}{a(t)}$

$1+z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{emit}}} = \frac{a(t_{\text{obs}})}{a(t_{\text{emit}})}$

Friedmann Equation Sphere with radius a & Newton: 1. rearrange to $kc^2/a^2 \rightarrow 2$. $E = K + V = \frac{1}{2} \dot{a}^2 - \frac{4}{3} \pi G \rho a^2$ factor out $H^2 \rightarrow 3$. define GR: $E \rightarrow -\frac{1}{2} kc^2 \Rightarrow \rho c = \frac{3H^2}{8\pi G} \Rightarrow$

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{a^2}$$

$\frac{kc^2}{a^2} = H^2(\Omega - 1)$ with $\Omega = \frac{\rho}{\rho_c}$

Note: **a, H, Ω, ρ** evolve with time. ($c dt = a d\eta$)

Scenario	k	Ω	$\dot{a}(t)$
Flat (slow down)	0	= 1	$a \cdot \sqrt{8\pi G \rho / 3}$
Closed (recollapse)	+1	> 1	$\eta = \pm \sqrt{\frac{da}{a \cdot \sqrt{\frac{8\pi G}{3c^2} \rho a^2 - k}}}$
Open (∞ -expand)	-1	< 1	

Procedure 1. define content $p(a) \rightarrow$ 2. insert $p(a)$ into Friedmann $\rightarrow$ 3. solve for $a(t)$ or $(a(\eta) \& t(\eta)$ if $k \neq 0$)

8.3 Contents of the Universe

1. Ideal Fluid $P = w \rho c^2$ with EoS parameter $w \rightarrow 2$. $U = \rho c^2 V_0 a^3$, $V = V_0 a^3 \rightarrow 3$. adiabatic exp. $dU + PdV = 0 \rightarrow$ 4. $\rho \propto a^{-3(1+w)}$ $\rho_i = \rho_{i,0} \left(\frac{a_0}{a(t)}\right)$ $\rho_{\text{tot}} \simeq \sum \rho_i$ $\rho_0^i = \Omega_i^0 \rho_{\text{crit}}^0$

Matter–Radiation Universe Early Universe: radiation-dominated until ($t \sim 5 \times 10^4 \text{ yr}$), then matter-dominated.

Non-rel. Gas, $i = M$
 $w = 0$, $\rho \propto a^{-3}$
today: $\Omega_{M,0} \simeq 0.31$
 $\Omega_{\text{Bary}} \simeq 0.045 \ll \Omega_{M,0}$

Radiation, $i = R$
 $w = \frac{1}{3}$, $\rho \propto a^{-4}$, $T \propto a^{-1}$
today: $\Omega_{R,0} \simeq 5 \cdot 10^{-5}$
CMB: $\rho_\gamma c^2 = a_B T^4$

Matter-Dominated Solutions $\Omega \equiv \Omega_{M,0}$, $\kappa \equiv |\Omega - 1|$

Flat ($k=0$, $\Omega=1$): $\rho_{\text{crit}} = \frac{3H_0^2}{8\pi G}$, $\frac{a(t)}{a_0} = \left(\frac{3}{2} H_0 t\right)^{2/3}$
Curved ($k \neq 0$): $\frac{a(t)}{a_0} / \Omega_0$, $\frac{a(t)}{H_0 t(n)}$
 $k=+1$ $\frac{2}{2n} (1 - \cos \eta)$ $\frac{\eta}{2n^{3/2}} (\eta - \sin \eta)$
 $k=-1$ $\frac{2}{2n} (\cosh \eta - 1)$ $\frac{\eta}{2n^{3/2}} (\sinh \eta - \eta)$
Early Epoch: $\frac{a(t)}{a_0} \simeq \left(\frac{3}{2} H_0 t\right)^{2/3}$
 $t_0 \simeq \frac{2}{3} H_0^{-1} \Omega_{M,0}^{-0.5}$ exact for $k=0$

Radiation-Dominated Solutions $\Omega \equiv \Omega_{R,0}$, $\kappa \equiv |\Omega - 1|$

Flat ($\Omega_R = 1$): $\frac{a(t)}{a_0} = \left(\frac{32\pi G \rho_{R,0}}{3}\right)^{\frac{1}{4}} \cdot t^{\frac{1}{2}}$
Radiation Temp.: $T \simeq 1.52 \times 10^{10} \left(\frac{1}{4\pi}\right)^{1/2} \text{ K} [E \simeq 1.31 \left(\frac{1}{4\pi}\right)^{1/2} \text{ MeV}]$ ($E = k_B T$ vs t)

8.4 Timescales

Energy	Time	Event
$E \gtrsim 10^{19} \text{ GeV}$	$t \lesssim 10^{-45} \text{ s}$	Planck Era
$E \sim 10^{13} \text{ GeV}$	$t \sim 10^{-34} \text{ s}$	Inflation
$E \sim 1 \text{ GeV}$	$t \sim 10^{-6} \text{ s}$	Quark-Hadron Equ.
$E \sim 1 \text{ MeV}$	$t \sim 1 - 100 \text{ s}$	Nucleosynthesis
$E \sim 10 \text{ eV}$	$t \sim 10^4 \text{ yr}$	R-M Equality
$E \sim 1 \text{ eV}$	$t \sim 10^5 \text{ yr}$	Recombination
$E \sim 0.01 \text{ eV}$	$t \sim 10^9 \text{ yr}$	Galaxies
$E \sim 1 \text{ meV}$	$t \sim 10^{10} \text{ yr}$	DE era